

OPEN SOURCE GUARANTEES POSITIVE CHANGE

The founder of an open source tricorder aims to make an accessible device that everybody can use to explore their environment, understand how things work, assess the levels of pollution, lighting, etc, and take positive steps to improve their area. Then, there is an open source underwater robot developed by a NASA engineer, which can be used by teachers to make studies on the ocean more interesting. Or how about an open source oximeter that can enable the setting up of a low-cost network to monitor blood oxygen levels of patients in a hospital? Or a pulsating pendant that simply makes you feel good? When we look at the purpose of many open source projects, their common aim seems to be to create some positive change – in terms of accessibility, discovery, learning, cost or innovation. And that is what makes open source so special!

A Bluetooth oximeter

<http://smartmaker.org/wiki/Projects:smARtPULSE>

Designed by Dimitri Albino, the smARtPULSE oximeter helps monitor blood oxygen levels, using standard photo-detection technology. A sensor is placed on a thin part of the patient's body, say a fingertip, and light of two different wavelengths is passed through the patient to a photo-detector. The changing absorbance at each of the wavelengths is measured, allowing the device to determine the absorbance due to the pulsing arterial blood alone, excluding vein blood, skin, bone, muscle, fat, etc. With near-infrared spectroscopy, it is possible to measure both oxyhaemoglobin and deoxygenated haemoglobin on a peripheral scale. The oximeter can connect to other devices using low-power Bluetooth 4.0, and people can use it out-of-the-box with the free app that runs on iOS and Android.

The open twist: What makes the smARtPULSE special is the complete open source nature of the project and the resulting ability to develop and manufacture the product at low cost, offer the best applications for tracking and analysing the readings, and ease of connecting to any computer running Linux, Windows or OS X. The team is working on a great application programming interface (API) and there will even be libraries available for Arduino, Raspberry Pi and Electric Imp. The special algorithm, hardware, protocol and API are all open source, so anyone can develop projects and

ideas based on the data received from smARtPULSE.

The team believes that it will help engineers connect the oximeter to the Internet of Things (Kevin Ashton's 1999-debuted concept of all objects being directly linked to the Internet via bar-codes or RFID, and without humans playing a role)—consider some examples like remote monitoring of health status and a low-cost network of oximeters for hospitals.

Tricorders that help observe, learn and discover

<http://www.tricorderproject.org/index.html>

Remember the tricorders used in Star Trek? A tricorder is a multi-function, handheld device used for sensor scanning, data analysis and recording data. With advancements in embedded systems and sensors, tricorders are becoming more of a reality. In fact, there are even contests such as Qualcomm's X-Prize that encourage engineers to make feature-packed tricorders. One such tricorder enthusiast, Dr Peter Jansen, has come up with a series of open source science tricorders code-named Mark. The tricorders provide a profound understanding of our environment by capturing varied information like temperature, humidity, atmospheric pressure, ambient light, distance and even magnetic fields. In his website, Dr Jansen speaks of how a tricorder can bring about positive change by offering a variety of information: "It's possible that the same instrument that can show a child how much chlorophyll a leaf has could also show how much pollution is in the air around us, or that's given off by one's car... If people could easily discover how their lifestyles impact the environment, they would then make positive lifestyle choices to reduce emissions, for instance. By having access to general tools, people can learn about leaves, air, clouds,

